

Development of an AI-based Surrogate Model for Real-Time Prediction of Stress and Deformation in Parametric CAE Systems.

1. Abstract

This project is to develop an AI surrogate model capable of predicting equivalent (Von Mises) stress and total deformation of a product in the form of a 3D model. It is done repeatedly for different parameters without performing computationally expensive finite element analysis (FEA). The study consists of a two-plate geometry (L-shape bracket) where there is a horizontal lower plate (LP) and a vertical upper plate (UP). Geometric parameters, including plate dimensions, fillet radius, and hole radius, are varied to generate a simulation dataset. Fixed support is applied to the hole on the upper plate while an external load is applied through the hole on the lower plate. Two machine learning models are investigated: Linear Regression for deformation prediction and Random Forest Regression for stress prediction. The Mean Absolute Error (MAE) is used to assess a model's performance. The results indicate that deformation has an approximately linear relationship with the input parameters, making Linear Regression a very suitable choice. Conversely, stress is nonlinear due to stress concentration around the fillet and hole, where Random Forest demonstrated better predictions. The proposed surrogate model can substantially decrease computational time while maintaining satisfactory prediction accuracy, thus showing its potential for rapid design evaluation and optimisation.

2. Introduction

(Finite Element Analysis) FEA is widely used in Mechanical Engineering to evaluate structural performance before physical manufacturing. However, performing a complete FEA for each design over many design iterations becomes computationally expensive during optimisation. AI provides an alternative solution to this using surrogate modelling. Machine learning is used to learn the relationship between design parameters and structural responses from simulation data. The trained model could predict these data at much higher speeds, making them attractive for real-time engineering applications. This project aims to develop machine learning models for predicting deformation and stress using a dataset generated from parametric FEA simulations.

3. Geometry & Modelling

The geometry consists of 2 plates forming an L-shape bracket. The lower plate (LP) represents the horizontal bottom plate, and the upper plate (UP) is the plate that's positioned vertically. The upper plate (UP) contains a fixed hole, while the lower plate (LP) contains a single loading hole through which the external force(1000N) is applied. Eight geometric parameters are varied: LP-L, LP-W, LP-T, UP-L, UP-W, UP-T, fillet radius, and hole radius (LP). These variables define the design space used for dataset generation.

(L-length, W- width, T-thickness).

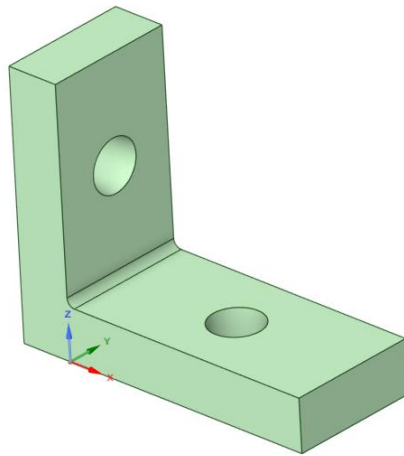


Figure 1: Parametric 3D model done using SpaceClaim.

Outline of Schematic B8: Parameters				
	A	B	C	D
1	ID	Parameter Name	Value	Unit
2	Input Parameters			
3	Static Structural (B1)			
4	P2	LP-L	100	mm
5	P7	LP-W	50	mm
6	P8	UP-L	100	mm
7	P12	LP-T	20	mm
8	P13	UP-T	20	mm
9	P14	UP-W	50	mm
10	P15	Fillet	3	mm
11	P16	Hole Radii	10	mm
*	New input parameter	New name	New expression	

Figure 2: Parameters

4. Finite Element Analysis

Finite Element Simulations were performed for each parameter combination. The upper plate hole was fixed while the lower plate hole had an external force of 1000N applied. A sufficiently refined mesh was used to capture stress concentration around the hole and fillet. The outputs extracted from every simulation were Equivalent (Von Mises) Stress and Total Deformation.

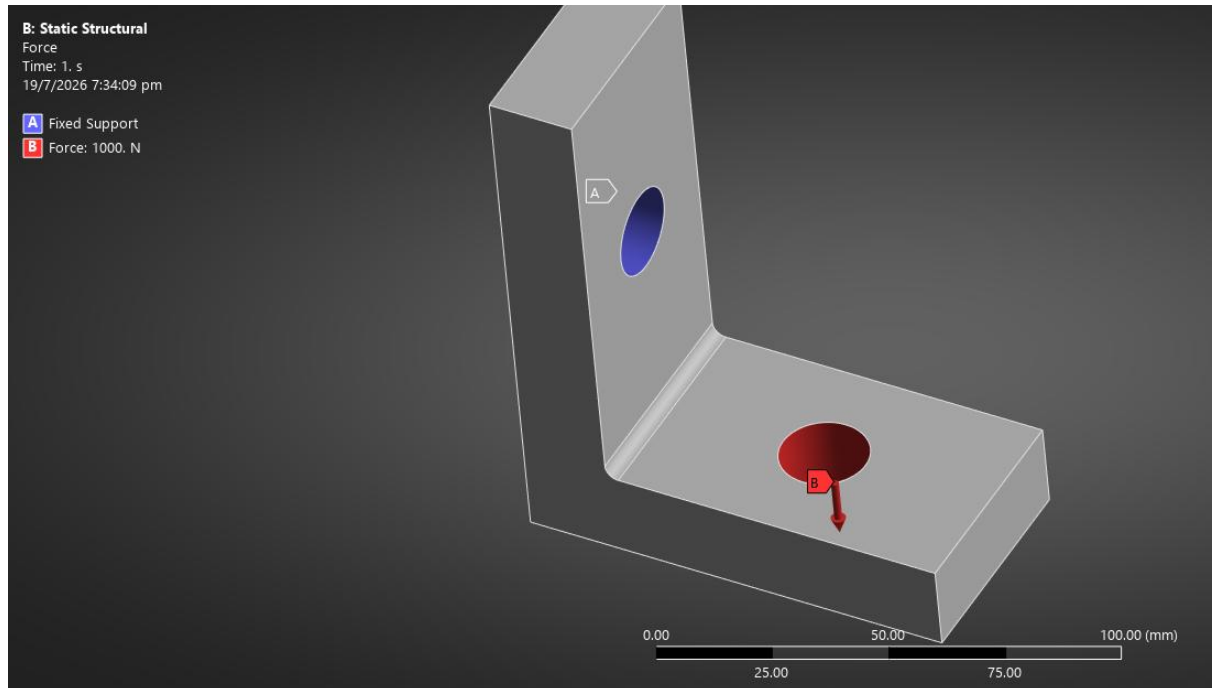


Figure 3: Boundary Conditions

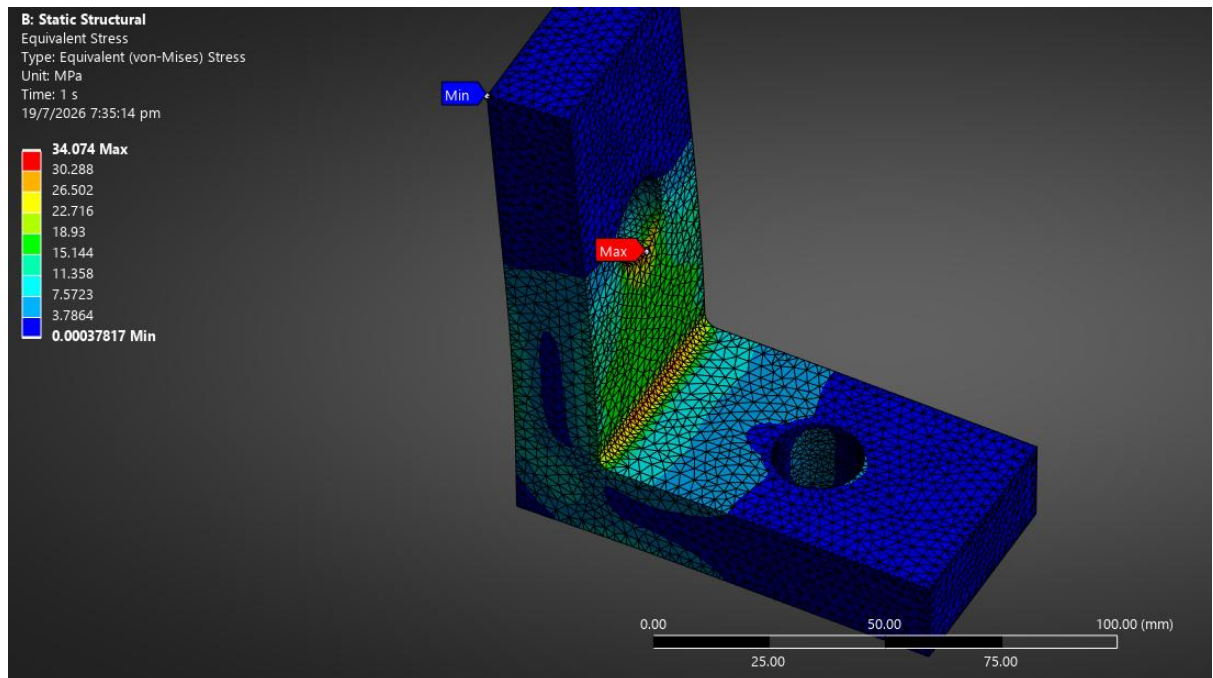


Figure 4: Stress distribution

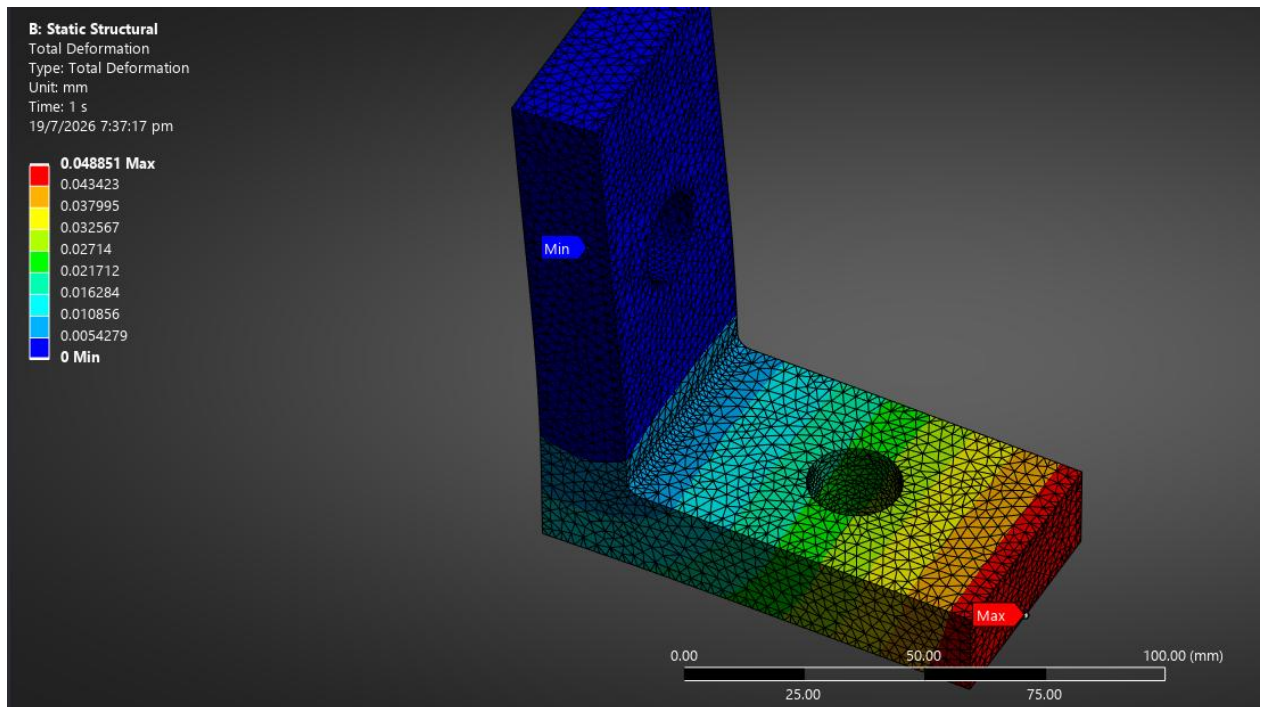


Figure 5: Deformation distribution

5. Dataset Generation

A parametric simulation workflow was made to generate multiple design configurations. Each row produces eight geometric parameters coupled with the stress and deformation values as the output values. This dataset forms the training and testing data for the machine learning models. The automated approach ensured consistent simulation settings while covering a broad design space.

1	Name	P2 - LP-L	P7 - LP-W	P8 - UP-L	P12 - LP-T	P13 - UP-T	P14 - UP-W	P15 - Fillet	P16 - Hole Radi	P17 - Equivalent Stress Maximum	P18 - Total Deformation Maximum
2	Units	mm	mm	mm	mm	mm	mm	mm	mm	MPa	mm
3	DP 0 (Current)	100	50	100	20	20	50	3	10	34.074	0.048851
4	DP 1	120	50	100	20	20	50	3	10	32.699	0.048853
5	DP 2	100	50	100	25	20	50	3	10	25.346	0.027809
6	DP 3	100	50	100	20	20	50	6	10	32.578	0.047192
7	DP 4	80	50	100	20	20	50	2	10	36.273	0.04936
8	DP 5	100	50	100	20	20	50	3	8	41.138	0.051095
9	DP 6	100	50	120	20	20	50	3	10	32.446	0.059156
10	DP 7	100	50	100	20	25	50	3	10	32.752	0.04067
11	DP 8	85	50	110	22	23	50	3	9	30.256	0.038048
12	DP 9	120	50	105	24	21	50	5	10	23.426	0.03012
13	DP 10	95	50	105	24	21	50	5	8	24.739	0.031444
14	DP 11	110	50	102	21	24	50	4	8	33.479	0.039109
15	DP 12	88	50	115	23	22	50	2	10	28.721	0.036568
16	DP 13	100	50	108	20	25	50	3	8	34.673	0.046274
17	DP 14	118	50	120	22	20	50	5	9	25.703	0.046453
18	DP 15	90	50	112	25	23	50	6	9	21.704	0.026783
19	DP 16	105	50	100	21	24	50	4	8	34.296	0.03835
20	DP 17	115	50	117	24	22	50	2	9	28.131	0.034124
21	DP 18	82	50	104	23	25	50	3	10	24.703	0.028953
22	DP 19	108	50	119	20	21	50	6	8	34.106	0.056746
23	DP 20	97	50	111	22	23	50	5	9	26.74	0.03756
24	DP 21	112	50	106	25	24	50	4	10	20.992	0.02427
25	DP 22	86	50	113	21	20	50	2	8	35.03	0.051731
26	DP 23	119	50	101	23	22	50	6	9	26.083	0.031159
27	DP 24	93	50	116	24	25	50	3	10	23.467	0.028913
28	DP 25	107	50	109	20	21	50	5	8	32.564	0.052442
29	DP 26	99	50	114	22	24	50	4	9	27.706	0.03762

Figure 6: Dataset Generation

6. Machine Learning Methodology

Two machine learning methods, linear regression and random forest, were used to obtain a more accurate model. Based on the results, the linear regression produced better accuracy for deformation. This is due to the deformation changing proportionally with the geometry under constant loading conditions, thus giving a better accuracy. Random Forest was selected for stress prediction. This is due to local stress concentration around holes and fillets, introducing nonlinear relationships. Model performance was evaluated using Mean Absolute Error (MAE). Predicted-versus-Actual graphs were generated to evaluate predictive performance. A coefficient plot was used to interpret Linear Regression, and feature importance analysis was used for Random Forest.

The coefficient plot in Figure 8 obtained from the linear regression model shows the influence of each geometric parameter on structural deformation. Parameters with large absolute coefficient values have a greater influence on the predicted deformation. Positive coefficients indicate that an increase in the parameter leads to higher deformation, while negative coefficients indicate a reduction in deformation. This provides interpretability to the model and helps identify the most critical design variables affecting structural performance.

The feature importance plot is done for the random forest model to predict the stress based on the contribution of each parameter. Random Forest captures complex nonlinear interactions between variables, making it more suitable for stress prediction. Figure 10 shows that certain parameters influence the stress distribution. This is consistent with engineering principles, where stress concentrations are significantly affected by geometry variations, particularly near discontinuities such as holes and fillets. Hence, this plot provides insights into which design parameter plays a key role in stress behaviour and design optimisation can be done based on that information.

7. Results

MAE (Mean Absolute Error)	Linear Regression	Random Forest
Stress (Mpa)	1.48	1.36
Deformation(mm)	0.00107	0.00266

Table 1

R-squared stress value for LR: 0.74

R-squared deformation value for LR: 0.98

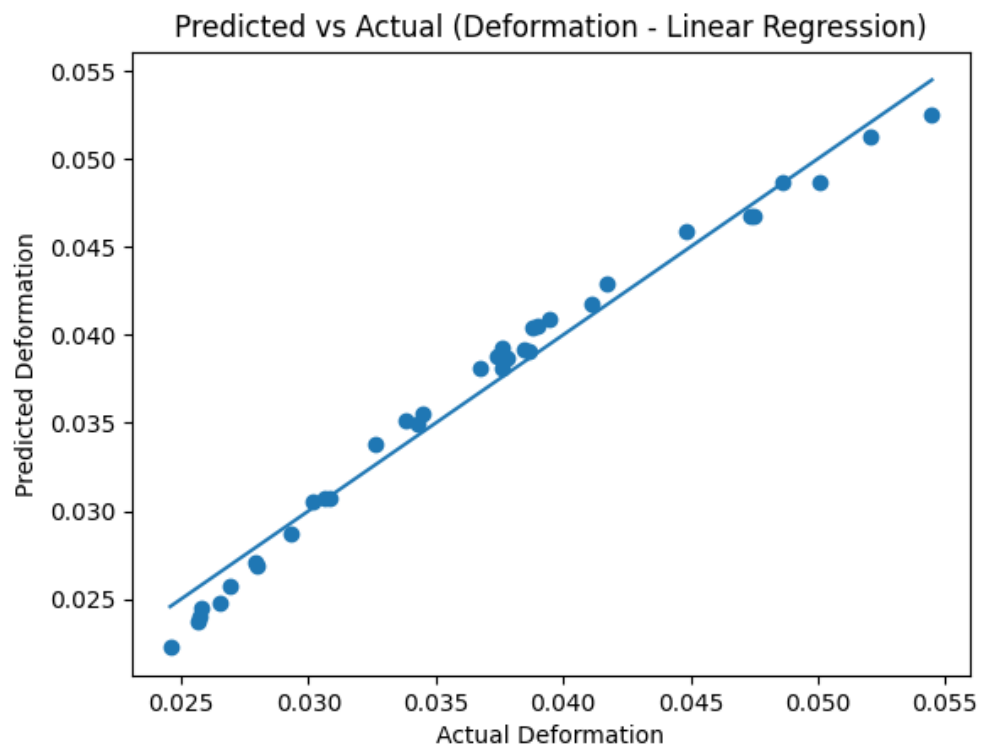


Figure 7: Predicted vs Actual deformation

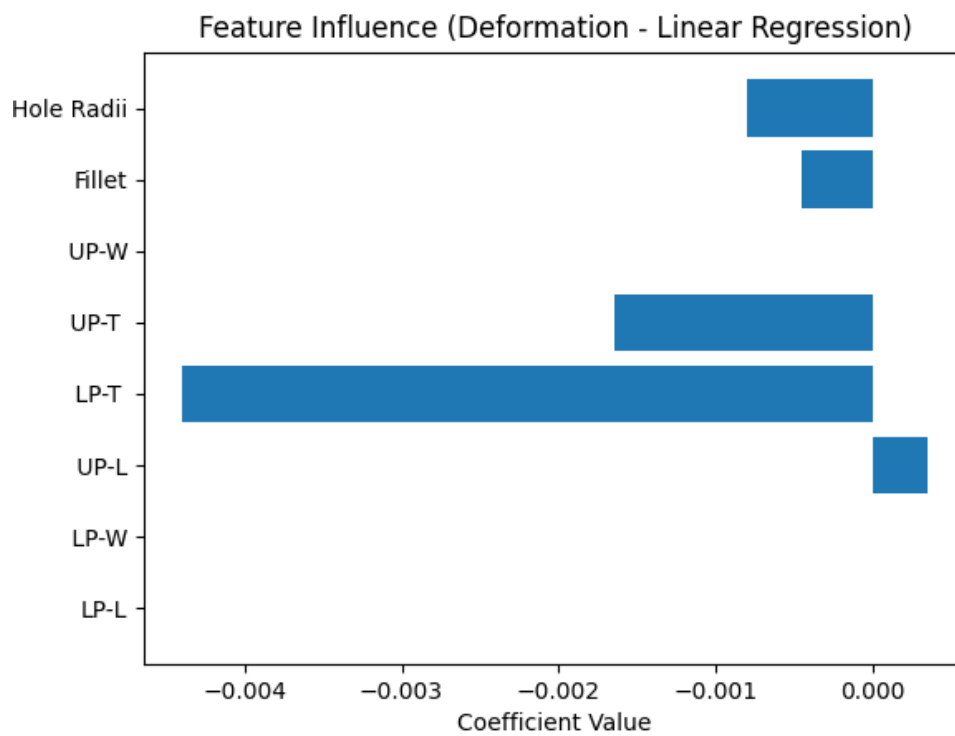


Figure 8: Linear regression coefficients

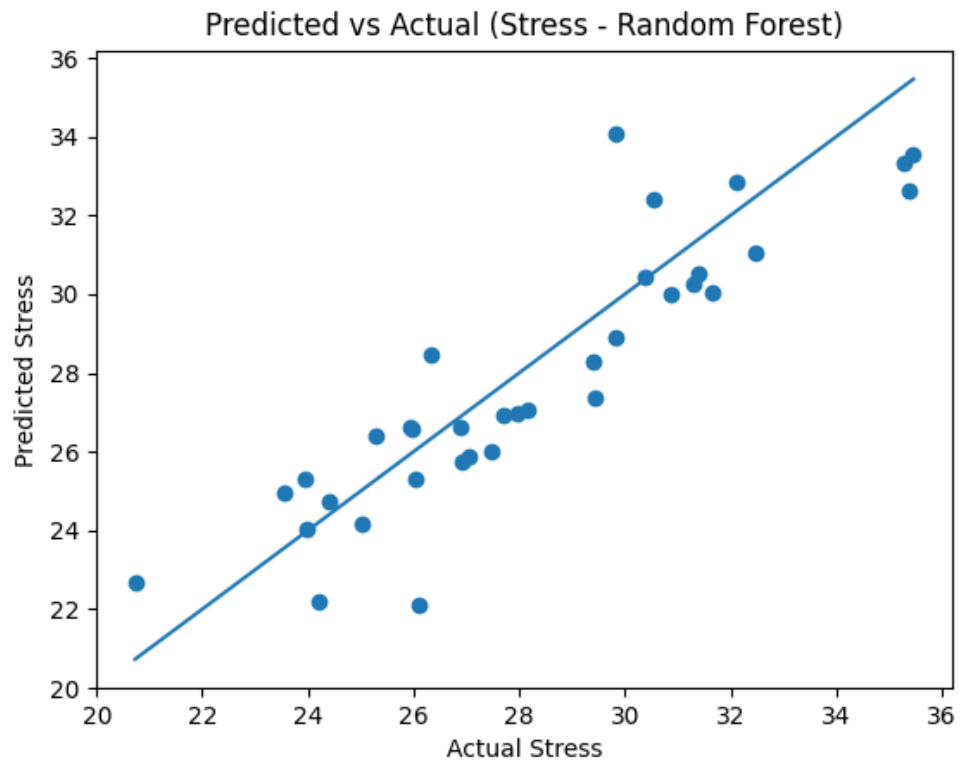


Figure 9: Predicted vs Actual Stress

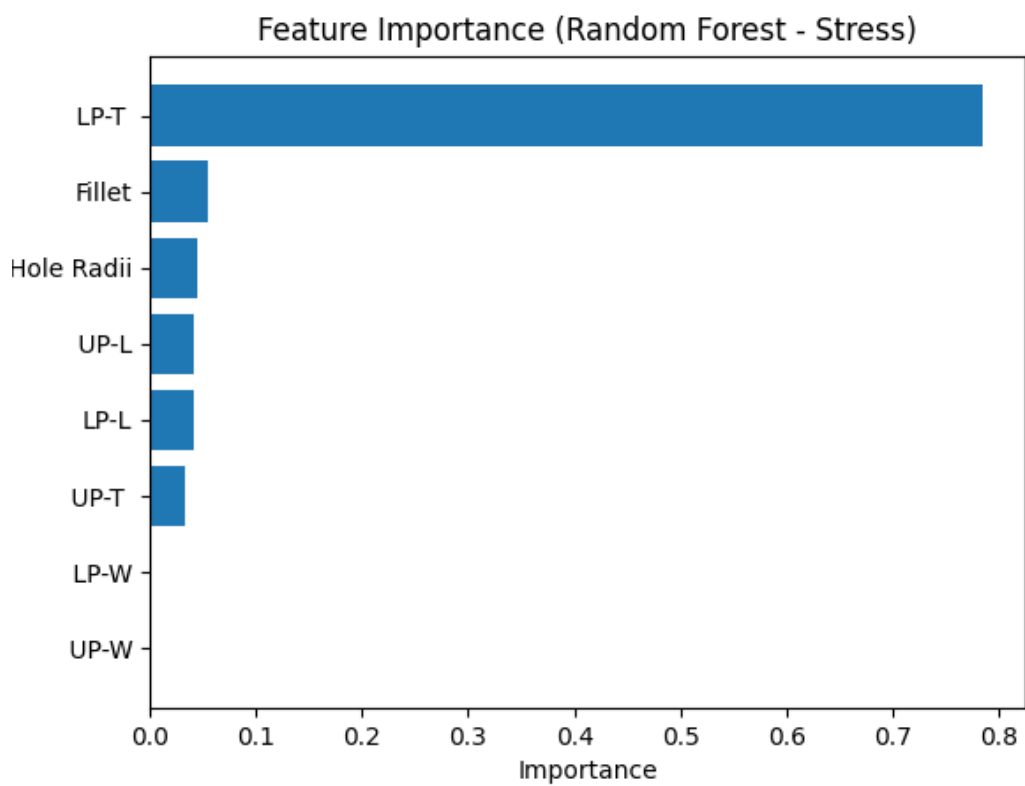


Figure 10: Random Forest Feature Importance

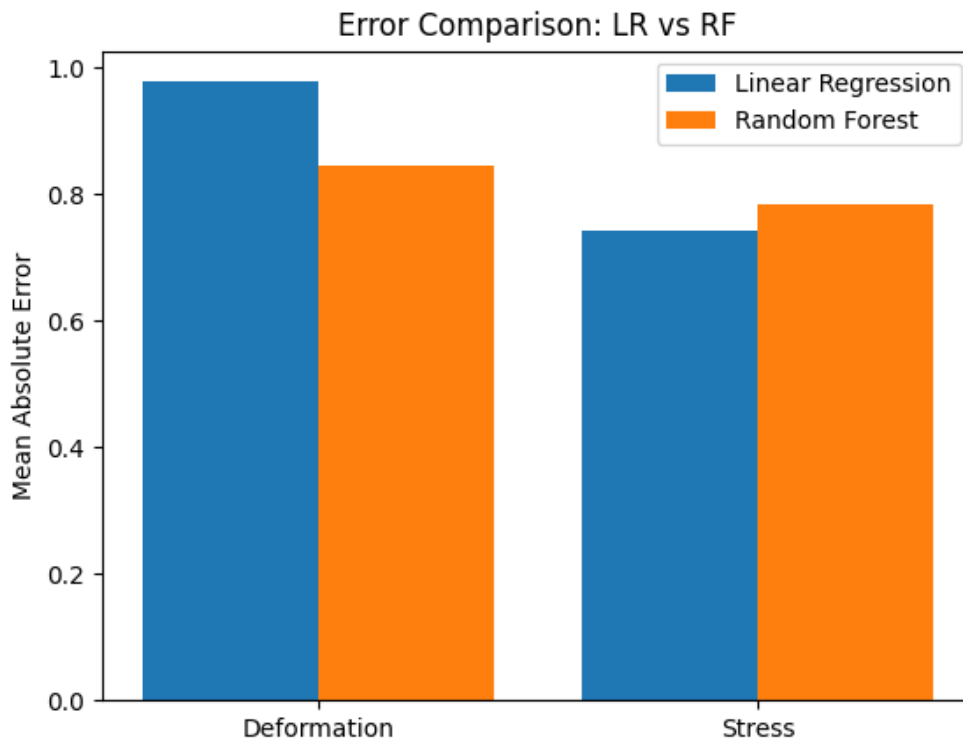


Figure 11: Error Comparison Plot

8. Discussion

The linear regression model achieved a very low prediction error, which confirms that deformation varies almost linearly with selected geometry parameters. The predicted values follow closely with the ideal values, indicating strong agreement with FEA results. The regression coefficient revealed how each geometric variable influences the deformation, and in this case, the lower plate thickness has the strongest influence. A high R-squared value was obtained for this, which supports the strong correlation between the actual and predicted values.

The Random Forest model produced superior stress predictions. Since stress is concentrated near holes and fillets, it has a non-linear characteristic. The Random Forest successfully captured these complex relationships, resulting in a lower MAE than a linear regression model. Feature importance analysis showed that hole radius, fillet radius and plate thickness had the strongest influence on stress.

Nevertheless, it's important to take note that even though 1.38 (MAE) seems small, it is important to consider the units used. Since the stress values are measured in MPa, even a small numerical deviation is a significant variation in real-world structural performance. This shows that even though Random Forest performs well, it still requires further improvement in accurately capturing the stress behaviour. In

engineering applications, these small errors can influence safety factors and design reliability, highlighting the importance and the need for further improvement.

Overall, the results show that selecting a machine learning algorithm according to the underlying physical behaviour improves predictive performance. The developed surrogate model gives quick results without relying on FEA, making it valuable for preliminary design optimisation and interactive CAD applications.

9. Conclusion

An AI surrogate modelling framework was successfully developed to predict the structural responses of an L-shaped bracket. The linear regression accurately predicted total deformation because deformation is approximately linear. The Random Forest effectively modelled stress distribution well due to its non-linear characteristics from stress concentrations. This model significantly reduces computation time and cost compared to repeating the FEA simulations while maintaining acceptable engineering accuracy. This study demonstrates that combining CAD, FEA and Machine learning supports faster design decisions in engineering.

10. Future Work

Further work includes increasing the dataset size, adding more geometric parameters and incorporating different loading conditions and material selection. This would reduce the stress MAE significantly. Further work includes exploring advanced machine learning methods such as Gradient Boosting, Artificial Neural Networks and Physics-Informed Neural Networks.

References

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- [4] Géron, A., Hands-On Machine Learning with Scikit-Learn, Keras & TensorFlow.